Assignment Title:

Predictive Modelling for Credit Card Default Using Machine Learning

Programme title:

MSc Artificial Intelligence

Module:

Introduction to Artificial Intelligence

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Year:

Term 1 – 2026

Submission Date:

10 / 06 / 2026

Issued by:

Dr. Abdelaziz Triki

Learner declaration

Item	Response
Word count	1424
Use of proof-reader/proof-reading service	NO
If yes, specify which service	N/A
I confirm that I submit this work as my own work and have cited all sources used	YES
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Introduction

Credit card default prediction is a key risk management problem for banks because a small share of customers who miss payments can generate a disproportionate share of losses. Machine learning models can use demographic, behavioural and credit variables to estimate default probability and support more informed lending and collections decisions.

In this project, a dataset of 34,788 credit card clients with 30 features was analysed to predict the target variable default.payment.next.month (1 = default, 0 = no default), as described by Yeh and Lien (2009). The data include demographic attributes (SEX, EDUCATION, MARRIAGE, AGE), credit limit and billing aggregates (LIMIT_BAL, BILL_AMT_SUM, LIMIT_BAL_LOG), and payment history (PAY0–PAY6), plus engineered variables such as CITY, RISK_RATING and a risk score risk_leak. The aim was to perform end-to-end exploratory data analysis (EDA), data preparation, model training, evaluation and interpretation.

The main contribution of the work is a comparative evaluation of six classifiers (Logistic Regression, Linear SVM with Platt scaling, Decision Tree, Random Forest, K-Nearest Neighbours and Gradient Boosting) on an imbalanced default dataset, with particular attention to feature importance and data leakage.

Exploratory Data Analysis (Task 1)

The target distribution is highly imbalanced: approximately 80.9% of clients did not default and 19.1% did default in the next month. This imbalance is clearly visible in the bar chart of the target variable and is consistent with real-world credit portfolios where defaults are rare events.

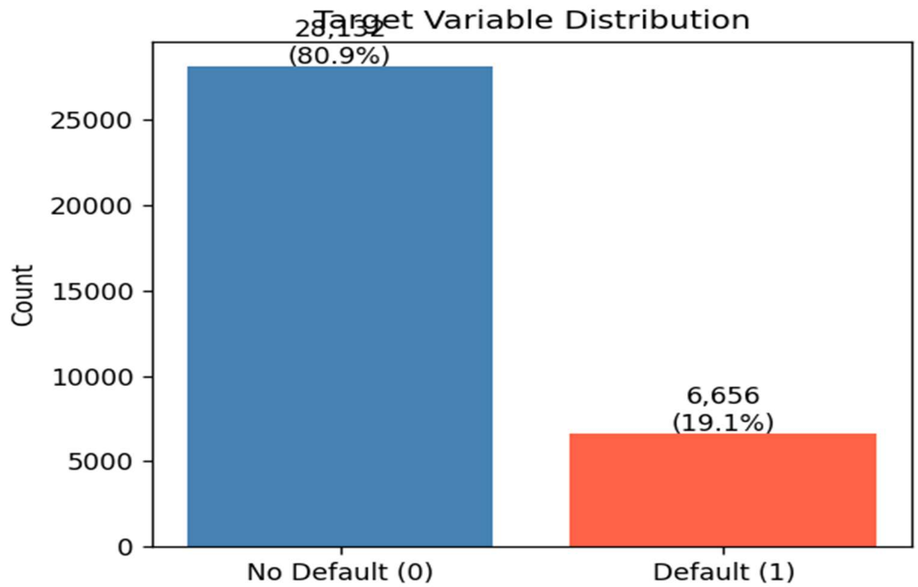

Figure 1: Distribution of the target variable (default vs. non default)

Descriptive statistics show that LIMIT_BAL, AGE, BILL_AMT_SUM and risk_leak are all skewed variables. BILL_AMT_SUM and AGE also show skewed distributions, which is consistent with long-tailed financial behaviour in the dataset.

Table 1: Descriptive statistics for key numerical features.

Feature	Mean	Median	Std Dev	Min	Max
LIMIT_BAL	184.7K	140.0K	235.9K	10.0K	5.6M
AGE	35.5	34.0	9.24	21.0	79.0
BILL_AMT_SUM	294.8K	129.0K	571.8K	-336.3K	21.3M
risk_leak	-56.9M	0.03	5.11B	-535.0B	412.0B

To visualise distributions, histograms and boxplots were generated for the main numerical features.

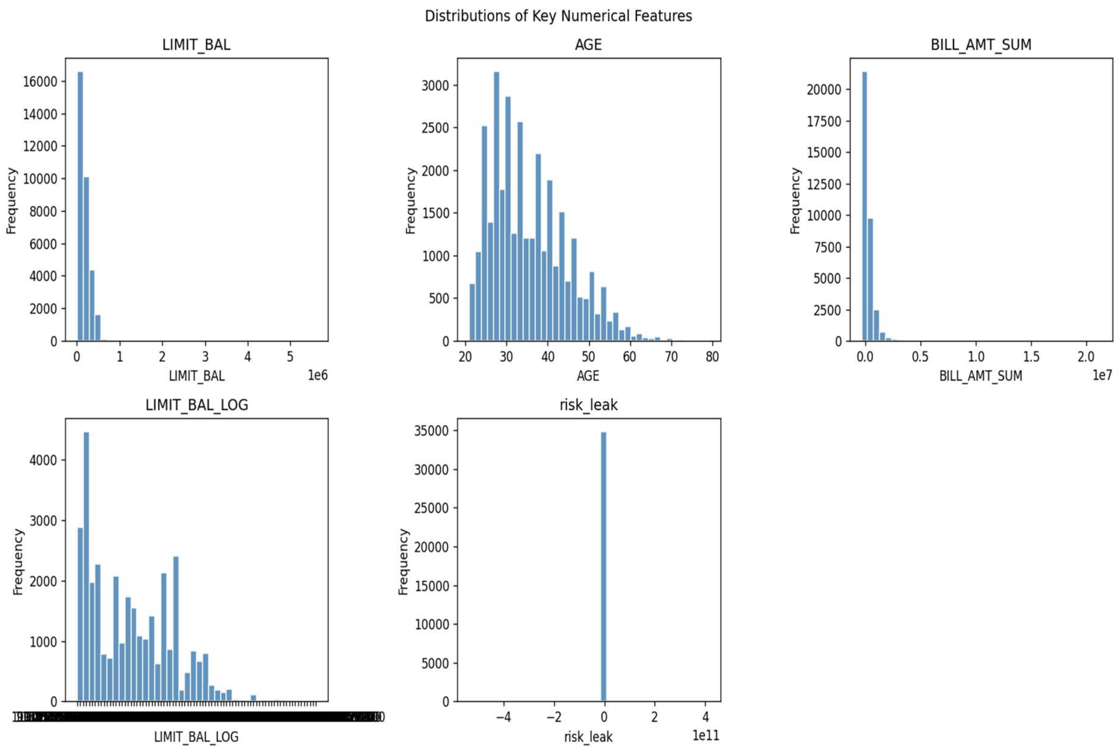

Figure 2: Histograms of key numerical features (LIMIT_BAL, AGE, BILL_AMT_SUM, LIMIT_BAL_LOG).

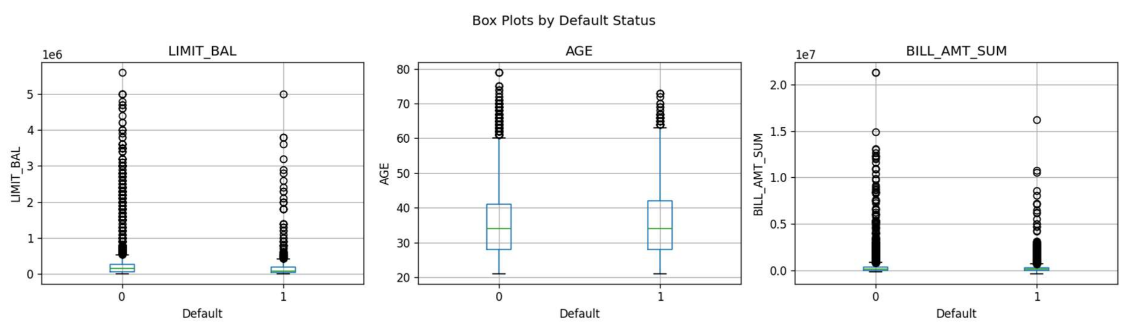

Figure 3: Boxplots of numerical features grouped by default status.

Categorical features show non-uniform distributions. The SEX, EDUCATION, MARRIAGE and RISK_RATING are clearly imbalanced across categories. CITY is high-cardinality, with activity concentrated in a small set of cities.

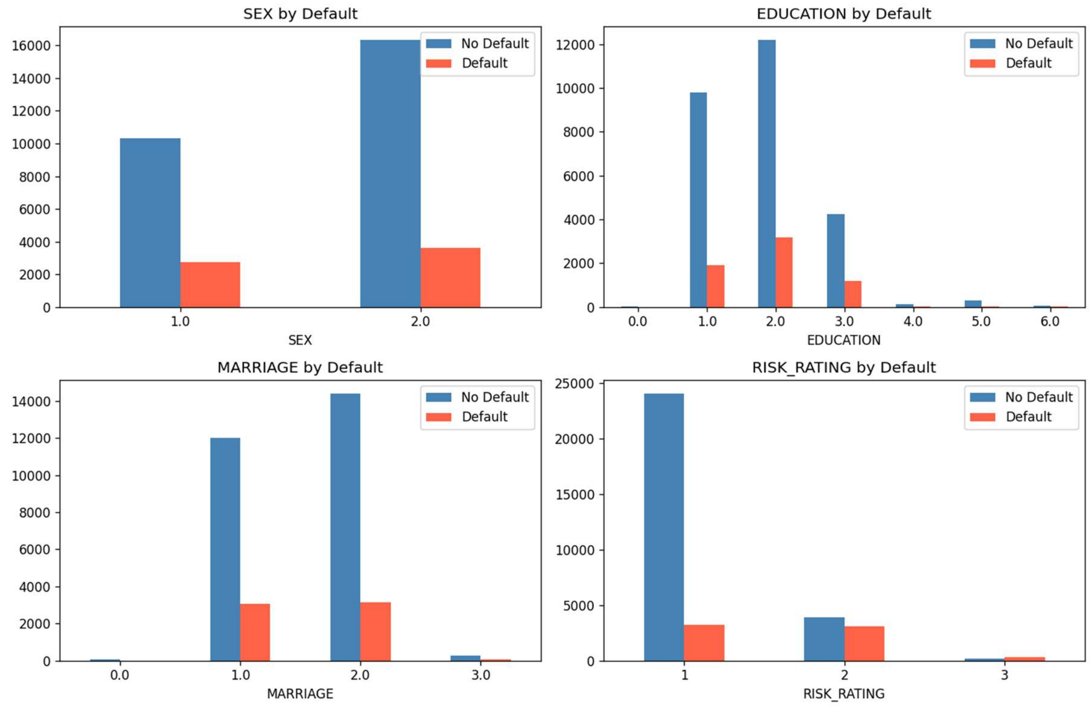

Figure 4: Distributions of categorical variables (SEX, EDUCATION, MARRIAGE, RISK_RATING).

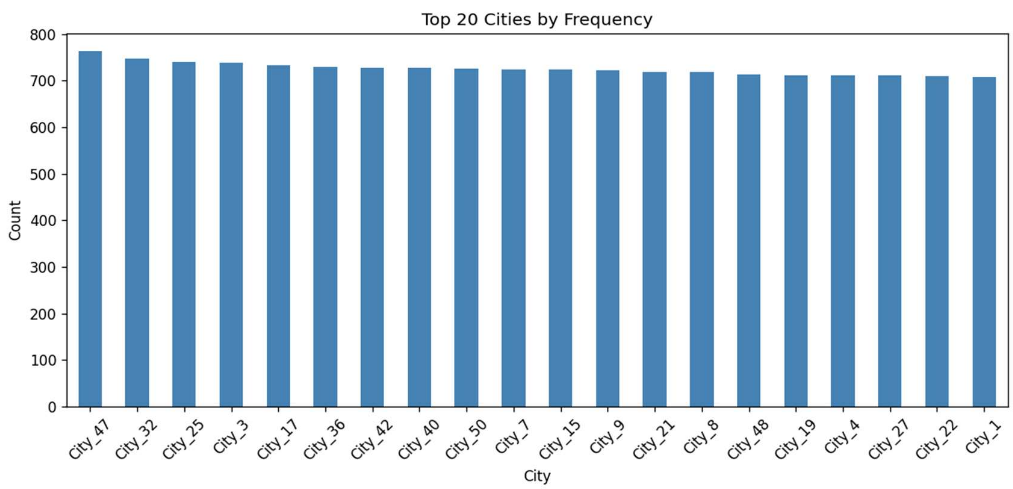

Figure 5: Top-20 city labels by number of credit card clients.

Correlation analysis indicates that payment history variables PAY0–PAY6, RISKRATING and billing/limit variables are most related to default risk. PAY0 (most recent delinquency status) in particular shows a strong association with the target. The engineered risk score risk_leak looked extremely predictive but was treated as a leakage feature and excluded from modelling (Kaufman et al., 2012).

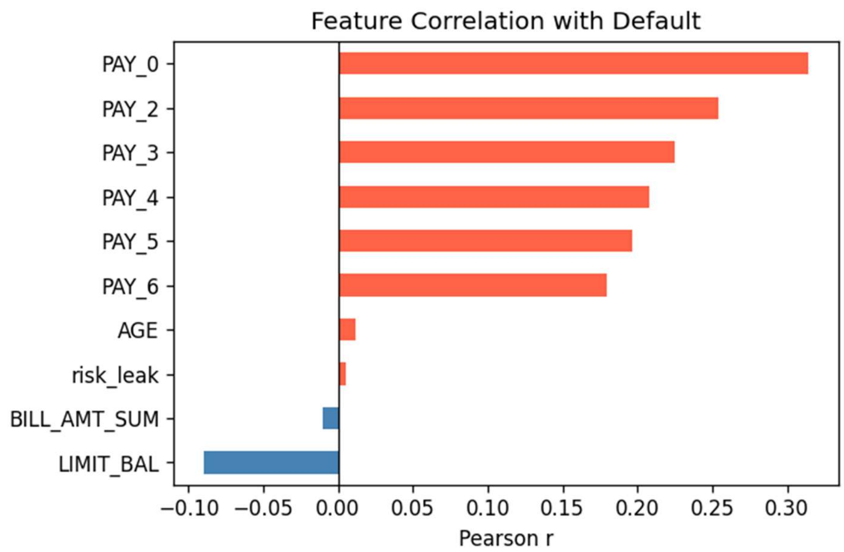

Figure 6: Correlation of selected features with the default indicator.

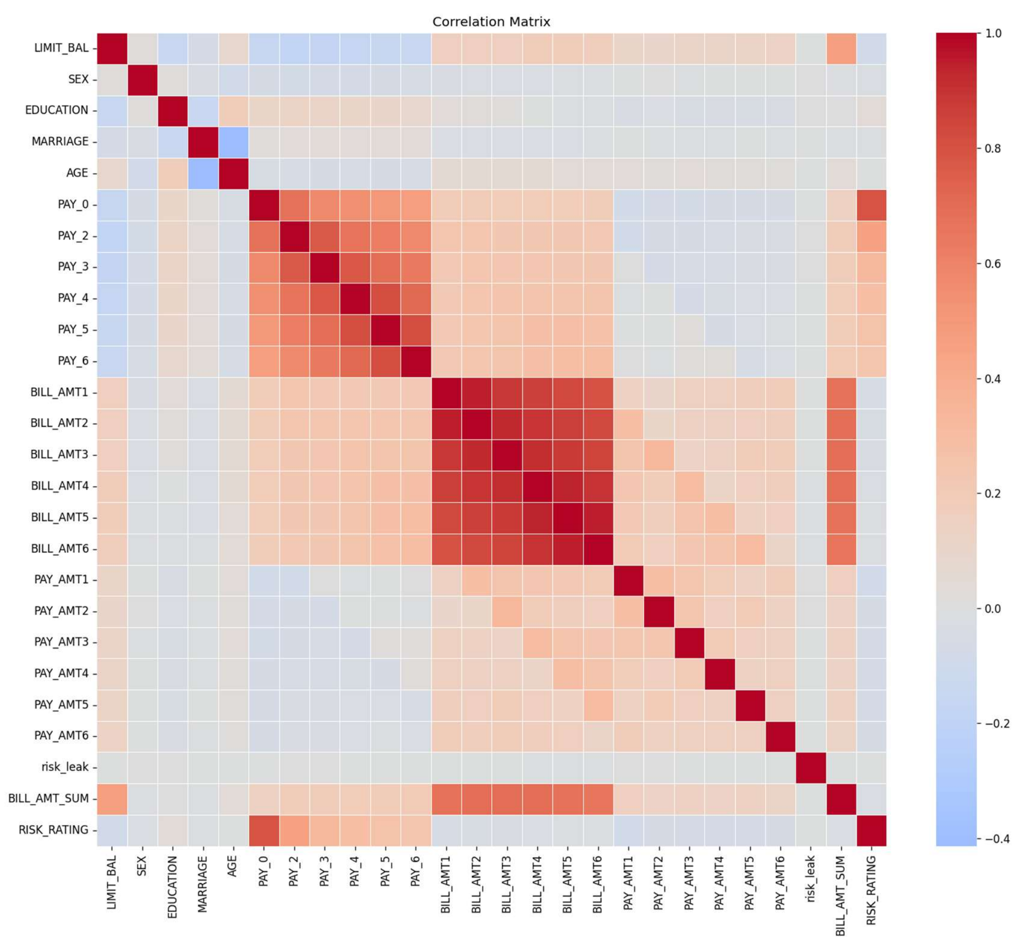

Figure 7: Correlation heatmap for numerical features.

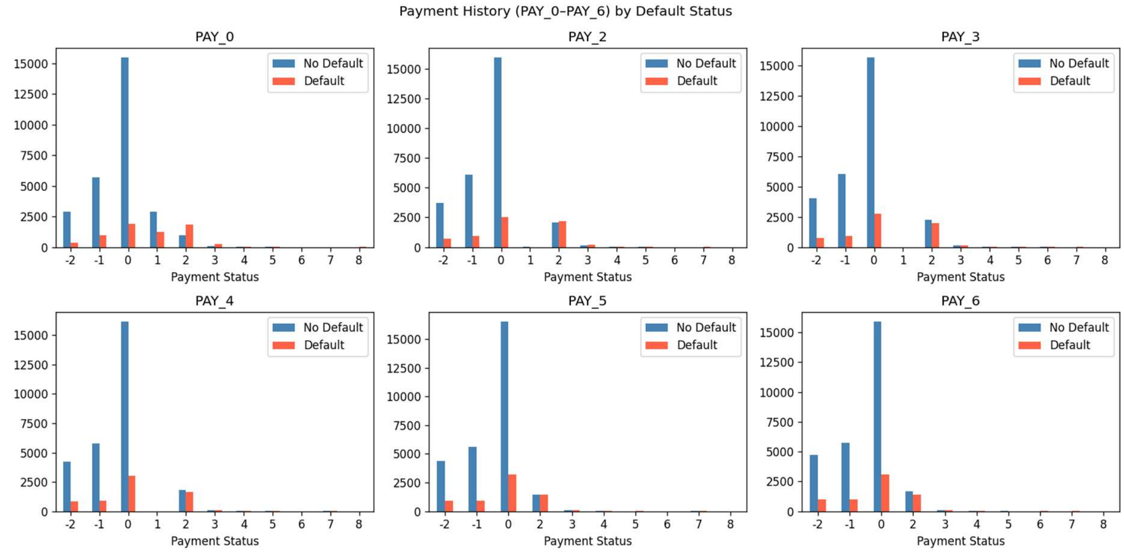

Figure 8: Payment history patterns for PAY0–PAY6.

Data Preparation (Task 2)

Missing values were first quantified across all columns and exported to missing_values_before.csv. LIMIT_BAL, several demographic fields, some payment amounts, LIMIT_BAL_LOG and risk-related fields contained missing values, while many others were complete.

Table 2: Missing values before preprocessing for selected features.

Feature	Missing count
LIMIT_BAL	1709
SEX	1767
EDUCATION	1734
MARRIAGE	1733
AGE	1755
PAY_AMT1	1725
PAY_AMT2	1744
LIMIT_BAL_LOG	1709

Median imputation was applied to all numerical columns to keep all records and handle skewed features such as LIMIT_BAL and BILL_AMT_SUM. After imputation, LIMIT_BAL_LOG was recomputed from the cleaned credit limit to ensure internal consistency between the raw and log-transformed features. The leakage-prone risk_leak and the ID column were dropped completely before modelling.

Because CITY had 50 categories, one-hot encoding would have created a very high-dimensional sparse matrix. Instead, CITY was label-encoded into an integer CITY_ENC, all remaining numeric features were scaled using StandardScaler, so that distance-based models such as K-Nearest Neighbours and linear margins (Logistic Regression and Linear SVM) operate on comparable feature ranges.

Finally, the data were split into train and test sets using an 80/20 stratified split, preserving the original 81:19 class ratio in both subsets. Stratification is recommended in the literature to obtain more reliable performance estimates on imbalanced classification problems. After preprocessing, no missing values remained and 27 predictors (excluding risk_leak) were used for model training.

Model Training (Task 3)

Six baseline classifiers were trained on the processed training set using default hyperparameters, recorded in default_hyperparameters.json for transparency and reproducibility. The models were:

- Logistic Regression with maximum 500 iterations and default regularisation.
- Linear SVM wrapped in CalibratedClassifierCV to obtain probability outputs for AUC computation using Platt scaling (Platt, 1999).
- Decision Tree with default depth and splitting criteria.
- Random Forest with 100 trees and two parallel jobs.
- K-Nearest Neighbours with 5 neighbours.
- Gradient Boosting with 100 estimators.

No resampling (e.g. SMOTE) was applied, so baseline models reflect the true class imbalance. All models were trained on an 80% stratified training split and evaluated on a 20% test set.

Model Evaluation and Visualisation (Task 4)

Each model was evaluated on the test set using accuracy, precision, recall, F1-score and AUC-ROC, and the results were exported to model_comparison.csv. A summary of baseline results (sorted by AUC-ROC) is:

Table 3: Baseline model performance on the test set (accuracy, precision, recall, F1, AUC-ROC).

Model	**Accuracy**	**Precision**	**Recall**	**F1**	**AUC-ROC**
Random Forest	0.8592	0.8318	0.3306	0.4731	0.8721
Gradient Boosting	0.8400	0.6501	0.3546	0.4589	0.7889
Logistic Regression	0.8224	0.5933	0.2269	0.3283	0.7501
Linear SVM (Platt)	0.8175	0.5556	0.2292	0.3245	0.7474
KNN	0.8275	0.5937	0.3118	0.4089	0.7248
Decision Tree	0.8353	0.6074	0.3929	0.4772	0.6664

Random Forest achieved the highest AUC-ROC and the best precision–recall trade-off, while Gradient Boosting also performed strongly, in line with credit-scoring studies showing that ensemble trees work well on imbalanced data (Brown and Mues, 2012).

Visual diagnostics included ROC curves and confusion matrices for all models.

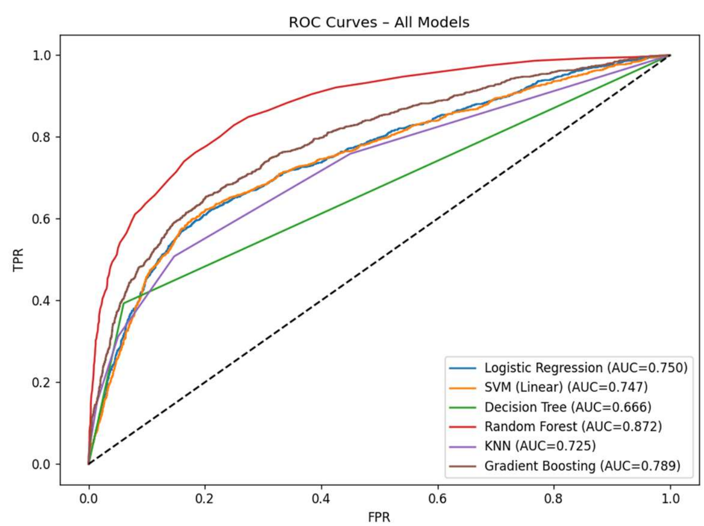

Figure 9: ROC curves for all classification models on the test set.

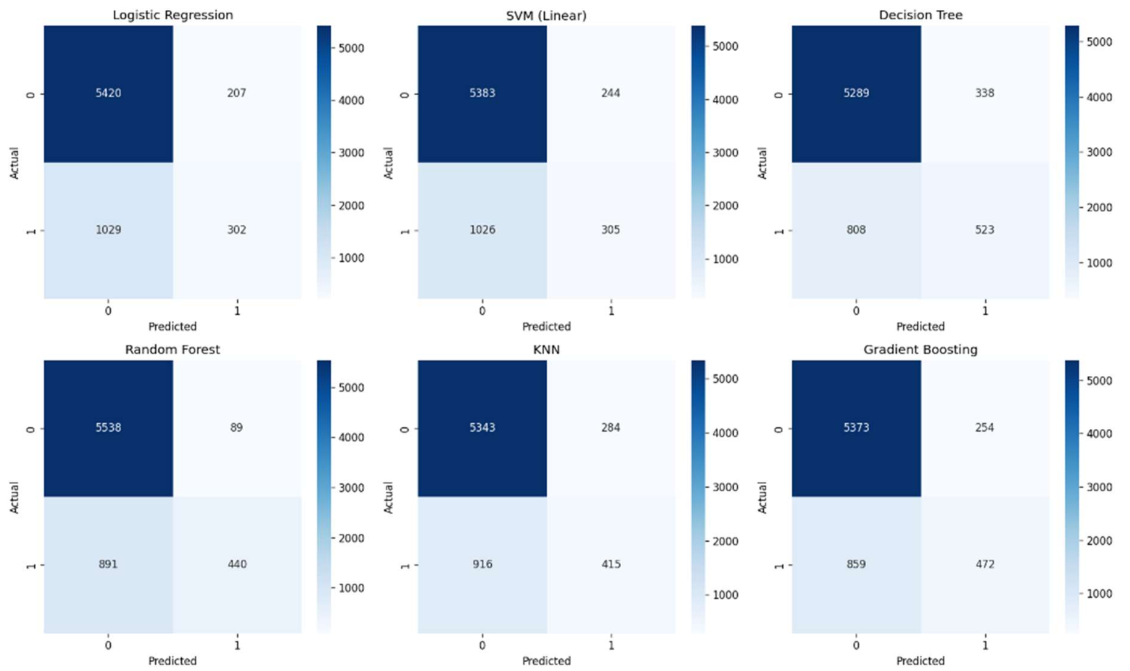

Figure 10: Confusion matrices for all classification models on the test set.

Confusion matrices show that models favor the majority class, so many defaulters are still missed. ROC curves confirm that Random Forest has the largest AUC and dominates other models across thresholds.

Hyperparameter tuning was then applied to the best baseline model. A grid search over Random Forest parameters (n_estimators, max_depth, min_samples_split) identified a tuned configuration of 200 trees, maximum depth 10, and minimum samples split 5. Tuned results were compared against the baseline Random Forest.

Table 4: Random Forest performance before and after hyperparameter tuning.

Version	**Accuracy**	**Precision**	**Recall**	**F1**	**AUC-ROC**
Baseline RF	0.8592	0.8318	0.3306	0.4731	0.8721
Tuned RF	0.8479	0.7191	0.3365	0.4585	0.7994

Interestingly, the tuned model slightly reduced AUC-ROC and F1, showing that naive tuning on one split can hurt performance.

Model Interpretation and Feature Importance

To interpret the best model, feature importance was analysed using permutation importance. Permutation importance on the tuned Random Forest indicated that PAY0 (current repayment status) was the most influential feature for distinguishing defaulters from non-defaulters. Other highly ranked predictors included RISKRATING, LIMIT_BAL, BILL_AMT_SUM and previous payment history variables (PAY2–PAY6), aligning with domain intuition that recent delinquencies and high utilisation drive default risk.

Table 5: Top features ranked by permutation importance for the Random Forest model.

Feature	Importance score
PAY0	0.032397997
RISK_RATING	0.017195215
LIMIT_BAL	0.00931354
BILL_AMT_SUM	0.010196785

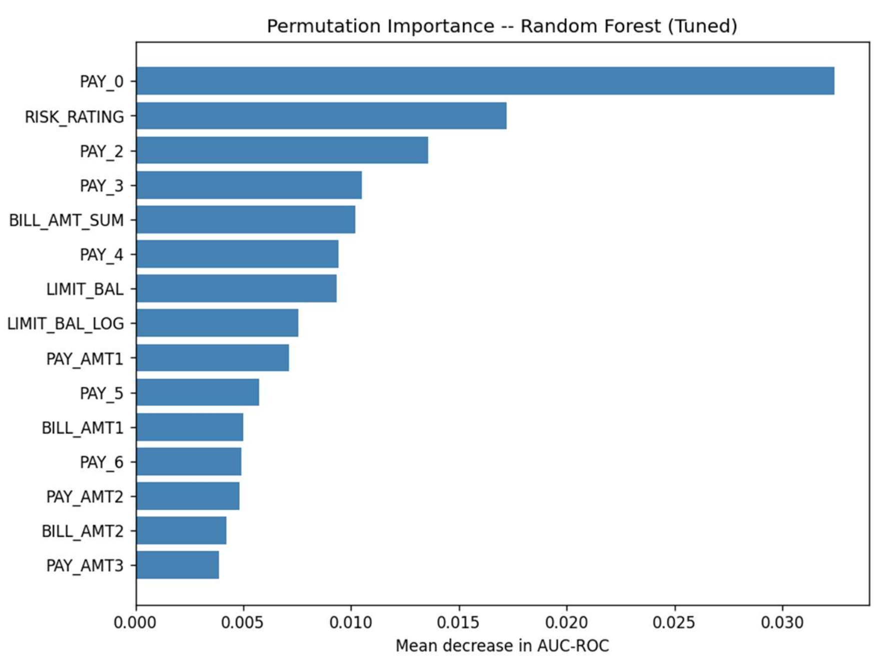

Figure 11: Feature importance for the Random Forest model (permutation importance).

A SHAP analysis was attempted but set aside due to shape mismatches in the calibrated model, so permutation importance is used as the main interpretation method.

Conclusion and Future Work (Task 5)

Across six classifiers, Random Forest achieved the best overall discrimination (AUC-ROC ≈ 0.87), followed by Gradient Boosting. Linear models and KNN/Decision Tree underperformed the ensembles, confirming that tree-based methods are well suited for this imbalanced credit default problem.

There are several limitations. First, the strong class imbalance means high accuracy hides modest recall on defaulters. Second, only a single train–test split was used. Third, fairness and demographic bias were not analysed.

Based on these limitations, three concrete directions for future work are:

1. Class-imbalance handling: Apply resampling techniques such as SMOTE or class-weighted loss functions, and compare their effect on recall and AUC-ROC, following recent studies on imbalanced credit scoring.
2. More robust validation and tuning: Replace the single 80/20 split with stratified k-fold cross-validation and perform hyperparameter tuning inside the cross-validation loop, as recommended by Kohavi (1995) for reliable model selection.
3. Fairness and deployment considerations: Analyse model performance across demographic subgroups (e.g. by SEX, AGE bands, MARITAL status) to detect potential bias, and design thresholding or monitoring strategies suitable for deployment in a real credit-risk environment.

Acknowledgement

Perplexity AI was used in the preparation of this report to support literature searching, identify potentially relevant academic sources, and assist with grammar correction and word-limit reduction during editing. All sources included in the reference list were checked by the author and cited as the original academic works rather than citing Perplexity AI as a source. The analysis, interpretation, discussion, and final written content submitted in this report are the author's own work and responsibility.

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